

Triple-Reduction Helical Gearbox Assembly Checklist

A. Gearbox information

Customer name	
Work order number	
Manufacturer	
Model	
Serial number	
Reduction ratio (overall)	: 1
Stage 1 ratio	: 1
Stage 2 ratio	: 1
Stage 3 ratio	: 1
Oil capacity (gal / L)	
Oil grade	

1. Prep and inspection

☐ Work order, drawings, BOM, and assembly spec on hand at current revision

☐ Clean covered work area, lifting gear rated and inspected

☐ Housing halves cleaned and deburred, oil galleries and drain/breather ports clear, no casting sand or chips

☐ Split-line faces flat, no nicks, dowel holes undamaged

☐ Bearing bores measured (diameter and roundness) against spec:

Spec / measured:

☐ Shafts checked for journal diameters, runout, keyway and thread condition

☐ Gears inspected for tooth damage, burrs, correct hand of helix, mating sets matched by serial/match marks

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- ☐ Bearings checked for part number, cups and cones kept as matched sets in packaging until use
- ☐ Seals, shims, gaskets, spacers, keys, fasteners counted against BOM
- ☐ Measuring tools calibrated (micrometers, bore gauges, dial indicators, torque wrench)

2. Shaft subassemblies (input, intermediate 1, intermediate 2, output)

- ☐ Keys fitted to keyways with correct fit
- ☐ Gears/pinions installed by press or shrink fit per spec (induction or oven, never open flame)
Heating temperature:
- ☐ Gear fully seated against shoulder, feeler gauge shows no gap
- ☐ Spacers and retaining rings/locknuts installed and locked
- ☐ Cones heated and installed (typically about 250°F/120°C max, confirm with bearing maker) and held against shoulder until cool
- ☐ Cone seating rechecked after cooling
- ☐ Runout of assembled gear checked:
Runout:

3. Housing preparation

- ☐ Cups installed square in bores (press or chill), fully seated against shoulder, confirmed with feeler gauge
- ☐ Seal bores clean, seals ready in correct orientation
- ☐ Internal surfaces painted or coated if specified

4. Shaft installation and gear mesh

- ☐ Shafts lowered in correct order (usually output, intermediate 2, intermediate 1, input, per drawing)
- ☐ Helix hands and thrust directions checked against drawing
- ☐ Mesh alignment confirmed, face widths centered, no axial offset beyond spec
- ☐ Housing closed with sealant or gasket as specified, dowels installed before bolts

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☐ Split-line bolts torqued in sequence:

Torque / sequence:

5. Bearing endplay or preload (each shaft)

☐ Bearing caps installed with starting shim pack

☐ Shaft rotated while seating to get rollers onto the cone rib

☐ Axial endplay measured with dial indicator (push-pull while rotating)

☐ Shims adjusted to target endplay or preload (record final shim thickness below):

Target:

☐ Rolling torque checked if preload specified:

Rolling torque:

☐ Cap bolts torqued:

Torque:

BEARING SHIM RECORD

Shaft	Location	Starting shim (in/mm)	Final shim (in/mm)	Measured endplay/preload	Spec	OK
Input	Drive end					☐
Input	Non-drive end					☐
Intermediate 1	Drive end					☐
Intermediate 1	Non-drive end					☐
Intermediate 2	Drive end					☐
Intermediate 2	Non-drive end					☐
Output	Drive end					☐
Output	Non-drive end					☐

Obtain all torque, preload, endplay, backlash, and clearance values from OEM drawings and bearing manufacturer data.

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6. Gear mesh verification

- ☐ Backlash measured at each stage with dial indicator at several points (record below)
- ☐ Contact pattern checked with marking compound (Prussian blue or gear marking paste) under light load, centered on tooth flank without heavy edge contact
- ☐ Shafts turn freely by hand, no binding through a full revolution of output

BACKLASH RECORD (measured at 4 points 90° apart)

Stage 1 Mesh: Input pinion to intermediate 1 gear

0°	90°	180°	270°	Min	Max	Spec	OK
							☐

Stage 2 Mesh: Intermediate 1 pinion to intermediate 2 gear

0°	90°	180°	270°	Min	Max	Spec	OK
							☐

Stage 3 Mesh: Intermediate 2 pinion to output gear

0°	90°	180°	270°	Min	Max	Spec	OK
							☐

Checked by:

Date:

7. Seals, closures, fittings

- ☐ Lip seals lubricated and installed with correct tool, lips undamaged and facing correct way
- ☐ Shaft surfaces under seals polished, keyway edges covered with sleeve to protect lip
- ☐ Inspection covers, breather, drain plug, sight glass or dipstick, oil lines installed
- ☐ All fasteners torqued and torque-marked

8. Lubrication and run-in

☐ Filled with specified oil grade and quantity:
Grade / quantity:

☐ No-load spin test: direction of rotation, speed, duration
Speed: Duration:

☐ Bearing temperatures monitored, stabilize below:
Temperature:

☐ Noise and vibration checked, readings recorded:
Readings:

☐ Leak check at seals, split line, plugs

☐ Contact pattern rechecked after run-in if required

9. Final and documentation

☐ Nameplate, rotation arrows, lifting points, oil fill tags attached

☐ Shaft ends protected, keys taped or secured, openings capped

☐ Serial numbers, bearing lot numbers, shim thicknesses, endplay, backlash, torques, test data recorded

Notes

Sign-off

Assembler

Name	Signature	Date

Inspector

Name	Signature	Date